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AI metrics and policymaking: assumptions and challenges in the shaping of AI

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Abstract
This paper explores the interplay between AI metrics and policymaking by examining the conceptual and methodological frameworks of global AI metrics and their alignment with National Artificial Intelligence Strategies (NAIS). Through topic modeling and qualitative content analysis, key thematic areas in NAIS are identified. The findings suggest a misalignment between the technical and economic focus of global AI metrics and the broader societal and ethical priorities emphasized in NAIS. This highlights the need to recalibrate AI evaluation frameworks to include ethical and other social considerations, aligning AI advancements with the United Nations Sustainable Development Goals (SDGs) for an inclusive, ethical, and sustainable future.

Keywords Artificial intelligence (AI) · AI metrics · AI policy · AI Governance · SDGs

1 Introduction

In today's complex and dynamic landscape, defined by the challenge of 'existential risks', informed and inclusive (Science and Technology) S&T policies are vital. Therein, measuring science and technology is pivotal in setting and determining realistic goals, aligning policy objectives with tangible outcomes (Debackere et al. 2019). Especially regarding artificial intelligence (AI), which is rapidly transforming almost all fields (Nature 2023). The metrics, however, that AI-related policies are based on are, as we argue here, problematic.

As AI becomes more integral to societal functions, the lack of appropriate metrics to evaluate critical dimensions of such integration can lead to significant oversight in S&T policy formulation and implementation. While metrics may effectively measure AI advancements in terms of innovation, talent competition and economy, they frequently overlook essential aspects like fairness, transparency, and privacy. Without robust metrics to gage fairness and equity, AI could exacerbate social disparities rather than ameliorate them (Mehrabi et al. 2021; Binns 2018). Lacking transparent algorithms, users and regulators cannot fully understand or trust AI decisions, potentially leading to breaches in accountability when errors occur (Ali et al. 2024; Konstantis et al. 2023; Ananny and Crawford 2018). Furthermore, as AI systems increasingly process large volumes of personal data, the respect for privacy remains a major concern that requires effective metrics to ensure AI systems uphold privacy standards and do not inadvertently become tools for surveillance (Golda et al. 2024; Wachter et al. 2017). Equally important is the role of metrics in assessing the impact of AI in advancing the United Nations Sustainable Development Goals (SDGs). Extensive research underscores the need for detailed and robust metrics to accurately evaluate AI's contributions to these goals. Without metrics that encompass global challenges, ethical and social dimensions, the effectiveness of AI in supporting critical SDGs, such as climate action, health and well-being and equity, cannot be fully realized (Liang et al. 2023; Rocchi et al. 2022; Chavarro et al. 2022; Gupta et al. 2021; Vinuesa et al. 2020).

Recent studies emphasize the urgent need to ensure that policies are in line with AI's transformative potential (Hine and Floridi 2024; Ward et al. 2024; Bengio et al. 2023; Nature 2020; Mishra et al. 2020). Current, however, policies heavily rely on inappropriate metrics, focusing on concepts like "competitiveness", "innovation" and "talent competition" (Erkkilä 2023). This should raise serious concerns as

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important AI policies inadequately consider indicators that highlight social and ethical considerations (Deloitte Insights 2023). Moreover, the challenge extends beyond metrics to the translation of value-driven social/ethical considerations into tangible policy actions, further emphasizing the need for recalibration. Policymakers are challenged to swiftly adapt their approaches to ensure alignment with AI's dynamic nature, addressing societal implications and potential biases inherent in the current metrics-driven policymaking paradigm (Schiff 2023).

Prompted by recent research advocating for responsible AI metrics (Sadek et al. 2024; Minkkinen et al. 2024; Papyshev and Yarime 2024) and related studies exploring nation-wide AI policy implications (Hine and Floridi 2024; Hälterlein 2024; Papyshev and Yarime 2023; Foffano et al. 2023; Saheb and Saheb 2023), our study seeks to delve into the intricate relationship between AI metrics and AI policy. To our knowledge, this is the first attempt to examine such an interplay. The objective of our own research is to explore how National Artificial Intelligence Strategies (hereafter, "NAIS") conceptualizes AI and examines whether the current AI metrics at a global level are consistent with these conceptualizations. Utilizing traditional topic modeling techniques (LDA) and qualitative content analysis for obtaining the topics of NAIS and the existing global AI metrics, this research sheds light on the intricate interplay of AI metrics and policy.

2 Methodology and data

This section outlines the methodological framework employed in this study, detailing the processes for data collection, identifying topics within NAIS, and determining global AI metrics. For a visual representation of the entire process, please refer to Fig. 3 in the Appendix.

2.1 Data collection

The first phase entails the identification and the corpus collection of all the National Artificial Intelligence Strategies (hereinafter "NAIS"). With an eye to constructing a comprehensive dataset, the following four official databases and platforms were probed into: the OECD AI Policy Observatory (OECD.AI 2024), the EU AI Watch Act (Van Roy et al. 2021), the STIP Compass (STIP Compass 2024), and the AI Governance Database by Nesta (2024). Triangulating the information from all databases, 53 countries have authored their NAIS policy documents. While processing the documents, we exclude several countries, namely Argentina, Chile, Colombia, Thailand, Indonesia, and Latvia from the analysis as the original documents' text was either not written in the English language or the identified documentation did not represent an official NAIS.[1] Hence, our final dataset consists of 43 documents, each representing a country's NAIS. The dataset includes information about the NAIS's country of origin, time span (issuance and expiration dates), and authorship (responsible governmental body).

The second phase pertains to the identification of existing metrics (indicators) relating to AI, within the context of S&T policy. Upon reviewing the available literature, it appears that there is currently no established national strategy that encompasses and publishes metrics related to AI. However, highlighting the importance of this aspect, certain institutional sectors have commenced the development of relevant initiatives (OECD 2024a), with dedicated research centers issuing specialized reports on AI metrics that are viewed as valuable resources to evidence-based policymaking (Perrault and Clark 2024). At the international level, novel global indices are crafted, specifically, for measuring AI: the Global AI Index (GAI), the Government AI Readiness Index (GAIRI), the Global Cities AI Readiness Index (GCAIRI), and the Artificial Intelligence and Democratic Values Index (AIDVI). All the above indices explicitly aim at analyzing countries' AI readiness in a meaningful way while AIDVI further focuses on evaluating progress toward trustworthy AI.

2.2 Topic delineation and content analysis

To delineate the topics discussed in the totality of the NAIS documents, we employ a mixed methods approach by combining qualitative content analysis with Latent Dirichlet Allocation (LDA) topic modeling technique (Blei et al. 2003). LDA topic modeling has been extensively used in literature, ranging from studying topic transition in social media platforms (Wang et al. 2012) to exploring topic evolutions within EU-funded research projects in social sciences (Kropp and Larsen 2023). It is a standard, unsupervised document classification technique that allows for an open interpretation of the obtained distribution of words. Qualitative content analysis complements LDA topic modeling by providing a nuanced understanding of the contextual and thematic dimensions embedded in the data (Schreier 2012).

Within the premises of this study, we conduct qualitative content analysis (coding and non-coding) approaches to complement and validate the LDA results, enabling a more comprehensive exploration and detailed description of the identified topics. This approach was deliberately chosen to address the known limitations of LDA's textual interpretation (Hagen 2018; Maier et al. 2021). The integration of qualitative and quantitative methodologies has proven

[1] For example, see here: https://ai-watch.ec.europa.eu/countries/switzerland/switzerland-ai-strategy-report_en

effective in policy research as demonstrated by Papyshev and Yarime (2023) and Isoaho et al. (2021).

2.3 LDA topic modeling

In the initial phase, as the documents primarily pertain to National Artificial Intelligence strategies, we anticipate a comprehensive exploration of AI across various facets of a nation's caveat. As such, documents were divided into distinct paragraphs for the purpose of conducting LDA. This process has previously been shown to outperform document-level analysis (Du et al. 2012; van Berkel et al. 2020). Second, the raw corpus of all NAIS documents was processed, performing all the necessary sequential steps involved in data preprocessing, such as tokenization, stopword removal,[2] lemmatization, and vectorization. To extract the distribution of topics, we used unigrams, bigrams, and trigrams, while we implemented the topic modeling within the Python (3.12.1) environment.

In the application of LDA to analyze document collection, hyperparameters alpha and eta were defined at values of 1 and 0.01, respectively. We chose this setting to accommodate a diverse array of topics within the corpus allowing for specific words or vocabulary that appear to be more significant. These parameters influence the distribution of topics across documents and the distribution of words within the topics, aiming to capture distinct thematic elements with clarity.

To ascertain the optimal number of topics for our analysis, we conducted a systematic optimization of the coherence score across a range of possible topic numbers from 2 to 20. The coherence score measures the semantic similarity among the top words in each topic, indicating how meaningfully the words are grouped. The optimization process revealed that the highest coherence score, recorded at 0.583, was achieved with a configuration of fifteen topics. This outcome suggests that seventeen (15) topics most effectively represent the semantic structures within the data. Following the identification of these topics, the top 20 words from each topic were extracted along with their respective weights, which signify the strength of association with the topic (for details please refer to the Data Availability section, 'topic distribution of words.csv').

2.4 Qualitative content analysis

To validate and refine the 15 topics identified by LDA, the authors used a complementary qualitative analysis approach using NVivo software to examine findings and delve deeper into the contextual characteristics of each topic. Nodes were created to represent the LDA topics, populated with the keywords associated with each topic based on their probability of occurrence (e.g., "talent", "skills", "program", "job" etc. for "Human Capital"). Relevant passages in the NAIS documents containing these keywords were identified and assigned to the respective topic nodes. Each passage was manually reviewed to ensure contextual accuracy, verifying that the keywords reflected the intended meaning of the topic rather than unrelated contexts. Generic concepts representing key themes were explored in greater depth and keywords with broad meanings, such as "innovation," "development," "ethics," "social," "infrastructure," "datum," and "investment," were carefully analyzed within their specific contexts. Through this process, codes were iteratively refined and grouped into broader, more cohesive categories, ensuring a deeper understanding of the thematic structure. While no additional topics emerged during this process, the analysis affirmed the comprehensiveness and reliability of the LDA-derived topics and offered insights into their description and functionality (see Findings section, Table 1 for the topics' description).

To identify which of the 15 topics were discussed in each NAIS document, we organized the documents by country, assigning each document to a case node. Using topic nodes and contextual keywords derived during the validation step, relevant passages were coded sequentially. The matrix coding query tool was then employed to create a matrix that highlighted the presence of each topic across countries, uncovering national patterns and variations in global AI strategies.

Both the identification of topics and their presence across all NAIS countries were triangulated and validated through a targeted, non-coding manual review of all 43 NAIS documents by the authors. This review ensured the appropriateness of the topic titles, the comprehensiveness of the topics discussed, and their contextual relevance within the policy framework.

2.5 Findings

2.5.1 AI topics

This section consists of two parts: the first part presents the topics identified through the application of LDA in tandem with the qualitative content analyses (both coding and non-coding) conducted across the entirety of the NAIS. It also highlights the presence and prominence of each topic across countries, uncovering variations in global AI strategies. The second part introduces the current global AI metrics and provides a critical analysis of the concepts they encompass. Finally, it evaluates their conceptual alignment with

[2] In addition, country names and subsequent adjectives related to origin (e.g., Spain, Spanish), standard abbreviations and disproportionally common terms such as "Artificial Intelligence", "AI" or "strategy" were removed from the documents' corpus.

Table 1 The topics discussed in the totality of NAIS documents

Topic	Title	Description
1	Research and Development (R&D)	Research and development, from an academic and industry perspective
2	Innovation Strategy	Framework for technological innovation, predominantly from a governmental standpoint
3	Public Sector	Administration and service with a focus on citizens
4	Private Sector	Business projects on development and investment prospects
5	Infrastructure	Technologies related to public infrastructure
6	Data Governance	Data, access and infrastructure
7	National Security	Defense AI strategies
8	National Challenges	AI-enabled solutions for societal issues (i.e., healthcare, agriculture, education)
9	Human Capital	Measures aimed at attracting and growing AI talent
10	Socioeconomic Risks	Diffusion, awareness, public trust
11	Ethical Framework	Non-enforceable and non-binding value-based principles (i.e., inclusivity, diversity, human rights)
12	Automation	AI-driven automation, especially on the labor market
13	Regulation	Legal frameworks, guidelines, and standards
14	Collaboration	Cooperation and alliances on an international level
15	Alignment with Transnational Organizations	Adherence to principles and recommendations by specific transnational organizations. (i.e., EU, OECD, UN)

the identified NAIS topics, presenting their methodological frameworks and data sources (see Table 3 in the Appendix section for further details).

The following table (Table 1) describes the topics identified and discussed in the totality of NAIS topics:

Most countries have authored their NAIS in 2019, with 14 countries publishing their strategies this year. Other notable years include 2020, with 11 countries, and 2021, with 8 countries authoring their NAIS. Observing the data, there is a small number of countries where the NAIS spans shorter periods, such as 3 to 4 years; for example, Italy (2022–2024) and Denmark (2019–2022). In contrast, most of the strategies last between 8 and 15 years. For instance, Malta's strategy spans from 2019 to 2030, and China's from 2017 to 2030. Noticeably, some countries have particularly long spans, such as Czech Republic (2019–2035). In other countries, such as Canada, Poland, Luxembourg, and Lithuania, only the date of the document's issuance is provided without mentioning the duration of the NAIS. By now, some countries (e.g., the USA, Germany, Norway, Austria, Estonia) have updated their NAIS to reflect more recent developments and to be aligned with their evolving national priorities.

It is to be noted that the majority of the topics identified in these strategies are compatible with the recommendations and principles of transnational organizations, such as the OECD, the European Union, and the United Nations. Such principles on AI stress values like inclusive growth, sustainable development, human-centered values, fairness, transparency, robustness, security, safety, and accountability, while focusing on ethical considerations, inclusivity, diversity and trust, as well as aligning AI with the Sustainable Development Goals (SDGs) to ensure it benefits humanity and advances global development (OECD 2024b; United Nations 2022, European Commission 2019).

In addition, a review of the existing literature reveals that most topics identified in our research correspond directly or conceptually with those explored in prior studies (Papyshev and Yarime 2023; Saheb and Saheb 2023; van Berkel et al. 2020). However, the topic that refers to the alignment of strategies with transnational organizations emerges as a distinct area of discussion ('Alignment with Transnational Organizations'). This is further corroborated through both quantitative and qualitative analyses, identifying it as a distinct topic that primarily reflects the European regions' focus on regulatory and cooperative frameworks, which are closely intertwined with both regional and international dynamics (see also Fig. 1). This distinction can likely be attributed to the broader scope of data and strategies analyzed in this study, as well as the refined preprocessing steps undertaken, which have enabled a more granular examination of these unique thematic dimensions.

2.6 Discussion of topics

To assess the distribution of covered topics across different countries, we perform a qualitative content analysis. This involves delving into the totality of the NAIS and evaluating whether each country has indeed established strategic objectives or implemented specific tasks related to each topic. The figure below (Fig. 1) reflects this analysis using a check mark to indicate countries that align with this criterion.

Country	Human Capital	R&D	Private Sector	Data Governance	Regulation	Innovation Strategy	Ethical Framework	Public Sector	Infrastructure	National Challenges	Socioeconomic Risks	Automation	Collaboration	Alignment with Transnational Organizations	National Security
United States	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓
United Kingdom	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓
Australia	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓			✓	✓
Austria	✓	✓		✓	✓	✓		✓	✓			✓	✓		
Canada	✓	✓	✓							✓					✓
China	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓
Czech Republic	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓			
Denmark	✓	✓	✓	✓	✓	✓	✓	✓				✓	✓		
Estonia	✓	✓	✓	✓		✓		✓	✓		✓	✓		✓	
Finland	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓		✓	✓	
France	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Germany	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	
India	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓		
Italy	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Japan	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	
Lithuania	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓			
Luxembourg	✓	✓	✓	✓	✓		✓	✓	✓	✓			✓	✓	
Malta	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	
Norway	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓	
Portugal	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Russia	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓
Serbia	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓				✓	
Singapore	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Spain	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Sweden	✓	✓	✓	✓	✓	✓	✓		✓	✓		✓			✓
Brazil	✓	✓	✓	✓	✓		✓	✓		✓	✓		✓	✓	
Bulgaria	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓	
Egypt	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			✓
Hungary	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓			✓
Ireland	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓			✓	✓
South Korea	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Mauritius	✓	✓	✓	✓	✓	✓	✓			✓	✓				
Mexico	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓			
Netherlands	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Peru	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓		✓	
Poland	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Saudi Arabia	✓	✓	✓	✓	✓				✓	✓					
Slovenia	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
Turkey	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		✓
United Arab Emirates	✓	✓	✓	✓	✓	✓		✓	✓	✓					
Uruguay	✓		✓	✓	✓		✓	✓			✓	✓			
Vietnam	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓		✓
Belgium	✓	✓	✓	✓	✓	✓	✓	✓			✓			✓	

Fig. 1 Distribution of topics in NAIS across countries (from most frequently to least frequently discussed)

Drawing from Fig. 1, the topics featured most frequently across National AI Strategies (NAIS) are: Human Capital, Research and Development (R&D), Private Sector, Data Governance, Regulation, Innovation Strategy, Ethical Framework, Public Sector, Infrastructure, National Challenges, Socioeconomic Risks, Automation, Collaboration, Alignment with Transnational Organizations, National Security.

Below, the 15 topics are presented along with their descriptions, complemented by insights derived from the word cloud in Fig. 2. For further information regarding the distribution of topics data, please refer to the link in the Data Availability section.

'Human Capital': This is the most prominent topic of NAIS, which relates to the development of AI talent through educational programs, training initiatives, and talent

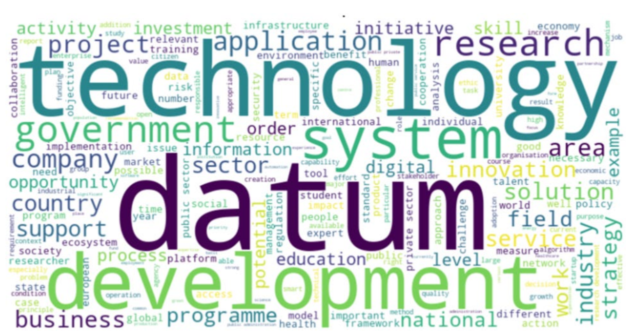

Fig. 2 Frequency of words (word cloud) in the totality of the NAIS documents

measures to nurture and attract a skilled workforce. This concept is also reflected in the word cloud through terms like "training", "skill", "talent" and "workforce" highlighting the focus on building human capacity for AI adoption and growth (Fig. 2).

'R&D': The second most discussed topic in NAIS is on Research and Development. It refers to funding schemes, research centers, and academia-industry collaboration to drive innovation and technological advancement. Terms like "research", "development", and "investment" have a high frequency in the word cloud underlining the significance of R&D in fostering AI-driven technological progress (Fig. 2).

'Private Sector': Another topic that receives equal attention is funding, accelerators, and technology support programs to encourage private companies in developing and applying AI solutions, alongside facilitating dialog while removing legal obstacles to foster innovation. Terms, such as "business", "industry", and "company", point to the role of private entities in shaping the AI landscape (Fig. 2).

'Data Governance': This topic relates to policies and mechanisms for data infrastructure development, including open data initiatives, data-sharing platforms, or regulatory frameworks that ensure responsible and ethical use of data in AI applications. Terms like "datum", "system", "platform", and "infrastructure" have a high frequency of occurrence signifying the critical role of data governance in enabling AI systems (Fig. 2).

'Regulation': Regulation of relevance to the development of legal frameworks, guidelines, and standards to regulate AI technology, including risk assessment mechanisms, transparency requirements and regulatory sandboxes to address societal impacts and ensure accountability. The word cloud terms "government", "regulation" and "standards" reflect the importance of this topic (Fig. 2).

'Innovation Strategy': This topic consists of implementing measures, such as pilot projects, innovation hubs, and procurement mechanisms, to stimulate innovation and adoption of AI technologies in the public sector. Terms, like "innovation", "initiative", and "strategy", highlight the relevance of this topic (Fig. 2).

'Ethical Framework': This topic includes the development of ethical principles, codes of ethics, and ethics committees to govern AI technology, with an emphasis on values, such as transparency, diversity and inclusivity, to address societal values and concerns. Terms like "value", "diversity" and "human" in the word cloud align with the ethical considerations embedded in NAIS (Fig. 2).

'Public Sector': This topic involves integrating AI technologies into public services through pilot projects, task forces, and awareness-raising initiatives, while optimizing public administration. The prominence of terms, such as "government", "technology", "public sector", and "service", in the word cloud underlines the strategic role of AI in transforming public services (Fig. 2).

'Infrastructure': This topic features the investment in digital infrastructure, such as high-performance computing, data centers, and network connectivity, to support AI development and deployment, with a focus on local language data resources in smaller countries. This may be reflected in terms, such as "infrastructure", "technology", "network", and "investment" (Fig. 2).

National Challenges: This topic embodies addressing societal issues like healthcare, agriculture, education, and smart cities through AI solutions, leveraging AI's potential to tackle pressing challenges and improve quality of life. Terms like "health", "education", "solution" and "field" in the word cloud reflect this focus on using AI to tackle pressing societal challenges (Fig. 2).

'Socioeconomic Risks': This topic comprises evaluating and mitigating risks associated with AI adoption, including awareness, trust, job displacement, inequality, and privacy concerns, through policies aimed at ensuring equitable access and distribution of AI benefits. This is reflected in the word cloud terms "trust", "risk", "awareness" and "public" (Fig. 2).

'Automation': The topic involves exploring the implications of AI-driven automation on various industries and employment sectors, with a focus on re-skilling and re-training initiatives to adapt to changing labor markets. The word cloud terms "process", "change", "application" and "job" depict the concept of automation on the workforce (Fig. 2).

'Collaboration': Among the less-discussed topics in NAIS, related to engaging in international cooperation, participating in policy discussions, and sharing expertise to foster a collaborative ecosystem for AI development. This is reflected in the word cloud terms "cooperation", "international", "collaboration", and "engagement" (Fig. 2).

'Alignment with Transnational Organizations': Another topic rarely touched upon in NAIS is that of aligning national AI strategies with the objectives and frameworks of transnational organizations like the European Commission, the OECD or the United Nations. The terms "order", "framework", "global", and "European" reflect this alignment of strategies with global frameworks (Fig. 2).

'National Security': The least-discussed topic is that of national security. It concerns addressing national security concerns through investments in defense AI, cybersecurity defense strategies, and military–civilian integration to safeguard critical infrastructure and protect against emerging threats. Terms like "security", "risk", "critical", and "national" underscore the relevance of this topic (Fig. 2).

2.7 AI metrics

In this section, we provide an overview of the available global AI indicators, exploring the dimensions underlying each of them. Additionally, we identify the key concepts associated with these dimensions and conceptually link them to the topics discussed in the NAIS. In total, four such indicators related to AI have been identified: the Global AI Index (GAI), the Global City AI Readiness Index (GCAIRI), the Government AI Readiness Index (GAIRI) and the Artificial Intelligence and Democratic Values Index (AIDVI).

The table below (Table 2) outlines the focus (subject), composition (dimensions), and the producers of the indices, with the conceptually aligned topics with the NAIS indicated in bold parentheses.

2.7.1 GAI

The Global AI Index (GAI) produced by Tortoise Media in 2019 assesses the national capacity for AI. It is updated in an annual manner and structured around three main conceptual pillars: *implementation*, *innovation* and *investment*. The *implementation* pillar specifically includes sub-pillars, such as "talent", "infrastructure", and "operating environment", with the first two aligning closely with the NAIS topics of 'Human Capital' and 'Infrastructure' (Tortoise Media 2024).

The "talent" sub-pillar draws on LinkedIn data to measure AI professionals, programming language downloads (R and Python from CRAN and Google), GitHub activity, and participation in AI-focused MOOCs (e.g., Amazon Alexa). It also uses Meetup and Kaggle data to gage the AI community's size and UNESCO data to count science, AI, and IT graduates. The "infrastructure" pillar combines generic technological indicators from organizations like the OECD and World Bank with IT-specific metrics from private sources (ISP Review, Top500, Speedtest).

As regards the "operating environment" sub-pillar, according to the defined methodology (Tortoise Media 2024), focuses on survey data that reflects public trust in Artificial Intelligence, diversity among practitioners, visa processing, and data governance as enabling factors. It evaluates factors like talent competition (visa policies), gender diversity (UNESCO, Kaggle), trust (Ipsos MORI), open data (OECD), and data protection (UNCTAD). This sub-pillar conceptually intersects with NAIS topics, such as 'Human Capital', 'Data Governance', 'Regulation', and 'Socioeconomic Risks'. However, when examining the more detailed "sub-sub-pillars" within the methodological framework used to evaluate the AI operating environment, it becomes apparent that these concepts/topics are not represented uniformly across the metrics.

For example, metrics that assess socio-economic risks or ethical aspects of AI such as trust or gender diversity in AI, as well as regulatory or governance aspects including level of data privacy legislation or participation in the international open data charter ('Regulation', 'Data Governance'), carry the least weight (overall weight < 0.6) compared to other "operating environment" indicators such as the cost of getting the fastest visa processing for high-skilled tech workers ('Human Capital'). The former carries a significantly higher weight; approximately three times greater (overall weight $= 1.5$). This discrepancy highlights a critical aspect of how different factors within the "operating environment" and GAI in general are prioritized. It points out that the weighting of indicators, such as trust in AI and gender diversity, is relatively low, which suggests a lesser focus on these ethical and socio-economic aspects. In contrast, more pragmatic factors like visa processing for high-skilled workers are given higher priority, reflecting a more immediate economic or operational concern for countries aiming to attract talent in the technology sector.

The *innovation* and *investment* pillars of the GAI focus on R&D and commercial ventures, incorporate university rankings (Times Higher Education, SCOPUS database), R&D spending, researcher counts (World Bank, UNESCO), and patents (GitHub, Google). Regulatory aspects are minimally represented but acknowledged in the "development" sub-pillar, which includes countries' participation in ISO AI standardization efforts. Both pillars mirror the NAIS's emphasis on fostering an environment conducive to technological advancement and economic growth through 'Innovation Strategy' and 'Private Sector' involvement.

Overall, the emphasis on NAIS's topics on 'Human Capital', 'Infrastructure', and 'Innovation Strategy' in GAI framework receives heavier consideration compared to the societal and ethical issues. This highlights a pragmatic focus where immediate economic benefits and enhancements in technological capacity are prioritized. These areas are critical as they directly impact a nation's ability to compete and innovate in the rapidly evolving global AI landscape. Conversely, the lighter weighting of ethical considerations, such as inclusivity and the socio-economic impacts of AI, suggests that these issues, while recognized, are not yet seen as immediately impactful to national competitiveness as the more tangible technology-driven factors. This realization might reflect an oversight in long-term strategic planning, where the socio-ethical dimensions of AI, critical for sustainable and equitable growth, are underemphasized.

2.7.2 GCAIRI

The Global Cities AI readiness index (GCAIRI) is updated annually and assesses the cities' AI readiness, is divided into four conceptual categories called "vectors", namely *vision*,

Table 2 Composition and structure of AI indices

Indices	Subject	Dimensions	Producer
GAI	National capacity for artificial intelligence	1.*Implementation* - Talent (**Human Capital**) - Infrastructure (**Infrastructure**) - Operating Environment (**Human Capital**, **Data Governance**, **Regulation**, **Socioeconomic Risks, Ethical Framework**) *2.Innovation* - Research, Development (**R&D**) *3. Investment* - Government strategy (**Innovation Strategy**) - Commercial ventures (**Private Sector**)	Tortoise Media
GCAIRI	Cities' AI readiness	*1.Vision* - Vision, Priorities, Mindset (**Innovation Strategy**) *2. Activation* - Quality of life and Diversity (**Human Capital**) - Demographic enablers (**Public Sector**) - Legal and governmental enablers (**Public Sector**, **Private Sector**) *3. Asset base* - Companies (**Private Sector**) - Workforce (**Human Capital**) - Funding (**R&D**, **Private Sector**) - Education and research (**Innovation Strategy**, **Human Capital**) - Infrastructure (**Infrastructure**) *4.Development and Trajectory* - Activation (development over time) (**Public Sector, Private Sector**) - Asset base (growth over time) (**Private Sector**, **R&D**, **Infrastructure**)	Oliver Wyman Forum
GAIRI	Government readiness to employ AI in public services	*1. Government* - Vision (**Innovation Strategy**) - Governance and Ethics (**Regulation**) - Digital capacity (**Innovation Strategy**, **R&D**, **Infrastructure**) - Adaptability (**Public Sector**) *2. Technology Sector* - Maturity (**Infrastructure**) - Innovation Capacity (**Innovation Strategy, Private Sector**) - Human Capital (**Human Capital**) *3. Data & Infrastructure* - Infrastructure (**Infrastructure**) - Data availability (**Data Governance**) - Data representativeness (**Public Sector**)	Oxford Insights
AIDVI	Assess progress toward trustworthy AI	*1. Frameworks for AI policy* - OECD/G20 AI Principles - UNESCO Recommendation on the Ethics of AI (**Alignment with Transnational Organizations**) *2. Human Rights* - Universal Declaration for Human Rights (**Ethical Framework**) *3. Democratic decision-making* - Public participation - Access to policy documents - Transparency (**Ethical Framework**, **Socioeconomic Risks)**	Center for AI and Digital Policy

activation, *asset base, development and trajectory* (Oliver Wyman Forum 2024).

The *vision* vector, specifically the “Vision, Priorities, Mindset” sub-pillar, evaluates how cities plan for technological advancements, reflecting the NAIS's emphasis on ‘Innovation Strategy’. This assessment underlines the strategic intentions of cities to harness AI technologies for future development. It further encompasses smart city plans, urban strategies, and economic development initiatives, supplemented by data from the World Economic Forum.

In the *activation* vector, the focus is on the attractiveness and governance of urban spaces through three sub-vectors: quality of life and diversity, demographic enablers, and legal and government enablers. The “quality of life and diversity” metric (Mercer and UN data), particularly its diversity aspect, evaluates urban inclusivity primarily through the lens of immigration policies aimed at attracting high-skilled individuals, thus highlighting NAIS’s topic of ‘Human Capital’ but with a limited scope on broader inclusivity measures. “Demographic enablers” address wealth distribution, linking to NAIS’s ‘Public Sector’ topic by considering societal equity. “Legal and government enablers”—building on the World Bank's Worldwide Governance Indicators and Doing Business rankings—signal existing rankings of good governance focusing on government effectiveness, ease of doing business and intellectual property rights connecting to ‘Public Sector’, ‘Private Sector’, as well as ‘Regulation’.

The objective of the *asset base* vector is to assess whether the city possesses the essential resources to realize its vision. The focal points are companies, workforce, funding, education and research, and infrastructure. Inquiring about whether a city possesses "a reservoir of talent in colleges and universities, an educated workforce, high-quality STEM education in primary and tertiary education, a track record for innovation and attracting pioneering companies, and the necessary infrastructure" (Oliver Wyman Forum 2024), aligns conceptually this dimension with the topics of ‘Human Capital’, ‘Research and Development (R&D)’, ‘Innovation Strategy’, ‘Private Sector’, and ‘Infrastructure’. Once more, the prominence of ‘Human Capital’, ‘Innovation Strategy’, and the ‘Private Sector’ stands out compared to other topics, giving this "vector" a resemblance to the concept of an "innovation ecosystem" while strongly reflecting the talent competition paradigm.

The "development and trajectory" vector, assessing changes in city capabilities over recent years, indicates a progressive outlook yet remains confined to traditional metrics of economic and governance effectiveness. It focuses on updates in government effectiveness, the business environment, venture capital flows, and infrastructure improvements, touching upon ‘Public Sector’, ‘Private Sector’ as well as ‘Research and Development (R&D)’ and ‘Infrastructure’.

Largely, the Global Cities AI Readiness Index (GCAIRI) offers a framework for assessing urban AI readiness, primarily emphasizing indicators centered around economy, innovation and human capital. However, this focus tends to sideline broader regulatory and ethical considerations essential for a sustainable AI ecosystem. In fact, the GCAIRI lacks any dedicated AI-specific metrics and relies entirely on preexisting indicators related to good governance, economic competitiveness, education, and innovation (Erkkilä 2023). This emphasis, while vital, may obscure the importance of inclusivity, regulation, and ethical frameworks, which are crucial for cultivating an ethical and socially responsible AI environment. To truly benefit society, AI global indicators should integrate a more holistic view that combines economic growth with ethical governance and social well-being (Foffano et al. 2023).

2.7.3 GAIRI

The Government AI readiness index (GAIRI) developed by Oxford Insights is updated annually and assesses the government readiness to employ AI in public services. It consists of three “pillars”: *government*, *technology sector*, and *data & infrastructure* (Oxford Insights 2024).

The *government* pillar covers the following four dimensions: vision, government and ethics, digital capacity, and adaptability. Similar to the GCAIRI, the vision dimension of GAIRI assesses the degree to which a country has formulated a strategic plan for technological innovation. This is primarily measured by the presence of a National AI Strategy (NAIS) and related initiatives. The data for this dimension are derived from sources, such as the OECD AI Policy Observatory and the UN IDIR AI policy portal, where countries’ published AI strategies are evaluated for comprehensiveness, scope, and innovation focus. The conceptual framework of this dimension is fundamentally tied to the NAIS topic that pertains to ‘Innovation Strategy’.

The government and ethics dimension encompasses metrics of data protection and privacy, cybersecurity, national ethics, and legal frameworks, aligning closely with the NAIS topic on ‘Regulation’. The digital capacity dimension includes metrics related to government-supported innovation initiatives promoting investment in emerging technologies, IT infrastructure, and online services, which are relevant to the topics of ‘Research and Development (R&D)’, ‘Innovation Strategy’, and ‘Infrastructure’. Data for this dimension are drawn from the UN e-Government Survey, the World Bank GovTech Maturity Index, and the Network Readiness Index. The adaptability dimension of the same pillar measures government effectiveness (sourced from the Worldwide Governance Indicators), responsiveness to change and e-procurement capacity (Global Data Barometer), features related to the ‘Public Sector’ topic. Similar to the development and trajectory dimension of GCAIRI, this dimension focuses on indicators that assess good governance.

The objective of the *technology sector* pillar is to assess the size of the sector that supplies governments with AI technologies (maturity) but also its “innovation capacity” and citizens' skill set (human capital). It's worth noting that aside from the count of AI unicorns, the metrics employed are general evaluations of the IT sector. Similarly, the innovation capacity dimension concentrates on the factors fostering innovation, encompassing entrepreneurial culture, and business administrative requirements. Such features mostly relate to the NAIS topics of ‘Innovation Strategy’, ‘Infrastructure’, ‘Private Sector’, and ‘Human Capital’.

The *data and infrastructure* pillar of GAIRI encompasses metrics concerning infrastructure, data availability, and data representativeness. Hence, this pillar is primarily associated with NAIS concepts concerning ‘Data Governance’ and ‘Public Sector’. Upon closer examination of the metric construction, it becomes evident that while the methodological approach regarding the infrastructure dimension is straightforward, the other two dimensions, albeit linked to AI,[3] lack clarity and comprehensiveness. Specifically, the notion of data availability for training AI models is confined to open government data and policies, statistical capacity, mobile phone and internet access coverage, thereby oversimplifying the concept of actual data generation, collection, and dissemination as merely dependent on internet access and ownership of digital devices. Similarly, data representativeness is evaluated solely based on gender disparities in internet access and socio-economic disparities in internet usage and the ability to acquire an internet-enabled device.

Overall, only 3 (three) of the 39 indicators in the Government AI Readiness Index (GAIRI) are specifically tailored to AI: (1) the presence of a national AI strategy (government pillar), (2) the number of AI Unicorns (technology sector pillar), and (3) the number of AI-related research publications (human capital pillar). While the index does address certain aspects relevant to AI policy, such as data availability for AI model training and data representativeness, these are treated superficially. Many metrics of GAIRI are adapted from conventional digital governance indices, which do not fully capture the unique challenges of AI. Furthermore, as highlighted by Erkkilä (2023), GAIRI's dependence on preexisting data underscores a continuity with traditional metrics that evaluate competitiveness, innovation, and digital governance. A comprehensive indicator intended to assess government readiness for AI should include factors that evaluate the ethical dimensions of AI, such as inclusivity, diversity, transparency, and human rights. It should also consider socio-economic factors like public awareness of AI and trust in its applications.

2.7.4 AIDVI

The "Artificial Intelligence and Democratic Values Index” (AIDVI) developed by the Center for AI and Digital Policy (CAIDP) is an annual assessment that focuses on promoting democratic values and human-centered policies in the development and use of artificial intelligence. AIDVI aims to evaluate countries based on their progress toward trustworthy AI, using criteria that emphasize transparency, fairness, accountability, and respect for fundamental rights. In total, 12 factors/questions are included in the methodological framework to assess national AI policies and practices. These factors are reflected in the following three dimensions (CAIDP 2024): 1) well-known *frameworks for AI policy* (the OECD/G20 AI Principles, UNESCO Recommendation on the Ethics of AI), 2) *human rights* (the Universal Declaration for Human Rights), and 3) *democratic decision-making* (transparency, public participation, and access to policy documents).

As regards the *frameworks for AI policy*, binary questions are employed to create metrics that assess the adoption and implementation of internationally recognized guidelines, such as the OECD AI Principles and the UNESCO Recommendation on the Ethics of AI. These questions are specifically designed to gage whether national AI strategies align with these global standards, as outlined in the NAIS topic: ‘Alignment with transnational organizations’. Though endorsement alone in such questions is not sufficient to determine country’s AI practices, such dimensions are in line with the objective to promote trustworthy and responsible artificial intelligence.

Similar questions are posed for the endorsement and implementation of the Universal Declaration for Human Rights[4] (‘Ethical Framework’). The authors note that although the Declaration predates AI, it is expected to underpin many upcoming policy debates due to its foundational principles.

In the third dimension of the index, which focuses on *democratic decision-making*, there is a notable emphasis on public participation in the development of AI policy. This dimension incorporates metrics and questions that evaluate the extent to which AI policies are broadly disseminated

[3] Issues pertaining to the data used to train AI algorithms, such as its availability and representativeness of the population, are crucial for mitigating biases and addressing concerns, such as fairness, transparency, accountability and other ethical considerations.

[4] The one notable exception is Saudi Arabia which did not endorse the UDHR but is a member of the United Nations and has recognized, according to human rights organizations, certain human rights obligations.

and how principles, such as algorithmic transparency, fairness, and accountability,[5] are promoted. These elements are crucial for ensuring that AI governance is transparent and inclusive. Moreover, these aspects are closely linked to NAIS topics concerning the 'Ethical Framework' and 'Socioeconomic Risks', underscoring their importance in shaping policies that govern AI.

Here, unlike the GAIRI and GCAIRI, AIDVI specifically includes AI-focused indicators, with 10 out of its 12 metrics directly related to AI aspects. However, the concept of trustworthy and responsible AI, although encapsulated by a set of critical yet broad questions, demands intensified scrutiny especially due to the rapid advancements in generative AI.

The development of a comprehensive and robust methodological framework for constructing AI metrics necessitates a multifaceted approach spanning from conceptual nuances of trustworthiness and responsibility of and in AI (OECD 2024a, b; Perrault and Clark 2024; Salloum 2024; Schoenherr et al. 2023) to the inclusion of important dedicated projects, databases, and initiatives that have been developed. Notably, the OECD AI Incidents Monitor,[6] which tracks AI-related incidents to concretely identify and mitigate AI risks, serves as a crucial tool for establishing trustworthy AI. Similarly, the AI Incidents Database[7] by the Partnership on AI catalogs failures of AI systems to enhance transparency and accountability. Such endeavors, coupled with significant academic contributions (Perrault and Clark 2024; Lin et al. 2021; Dhamala et al. 2021; Gehman et al. 2020) are essential in developing evaluation frameworks that comprehensively capture the intricacies of AI technologies and measure their trustworthiness and responsibility. This approach not only highlights the ongoing need for rigorous oversight mechanisms but also underlines the importance of global cooperation in developing standards that can guide the ethical deployment of AI systems worldwide.

3 Discussion and implications

By examining the conceptual and methodological framework of current global AI metrics and the way that AI is conceptualized in National Artificial Intelligence Strategies (NAIS), this research reveals a critical misalignment between the two. Despite the mentioning of ethical frameworks and social considerations of AI within NAIS, global AI metrics largely overlook these aspects. While global AI metrics heavily monitor the technical and economic aspects of AI, ethical and other social dimensions, such as the impact on public services, legal and ethical adherence, and national sector-specific challenges, are significantly underrepresented.

Highlighting this gap, this study suggests that AI metrics, in their current form, may shape AI policies that prioritize technological innovation over societal and ethical concerns. This realization is critical, as what is being measured (and how) significantly shapes the development of policies, which in turn influences societal functions (Muller 2018; Beer 2016; Espeland and Stevens 1998). The findings support the growing argument that responsible AI governance demands metrics that integrate both technical aspects and ethical concerns, as emphasized in recent literature (Minkkinen et al. 2024; Papyshev and Yarime 2024; Schiff 2023; Erkkilä 2023). Given the uncertainty surrounding AI's rapid evolution (Nordström 2022), such metrics must remain adaptable to both technological and societal changes and regularly monitored within a dynamic AI policy framework.

'Human Capital' is the topic that is prominently featured in almost all global indices. Recognizing the importance of talent and skills necessary for AI development, these indices evaluate educational frameworks, workforce capabilities, and the availability of skilled professionals in the field of AI, thereby reflecting a focus on enhancing the workforce to meet the demands of AI-driven economies (Papyshev and Yarime 2023; Saheb and Saheb 2023; van Berkel et al. 2020). Topics, such as 'Innovation Strategy', 'Private Sector', 'Infrastructure' and 'Research and Development (R&D)', receive substantial attention across most global AI indices like the Global AI Index, the Government AI Readiness Index and the Global Cities AI Readiness Index (Schiff 2023; Erkkilä 2023). These indices assess national capacity for AI, focusing on aspects like technological infrastructure, investment in AI technologies, and innovation capacities, which are crucial for a country's competitiveness in AI.

On the other hand, topics concerning ethical and societal dimensions of AI, such as 'Public Sector', 'Regulation', 'Data Governance', 'Ethical Framework', 'Socioeconomic Risks' and 'Alignment with Transnational Organizations', are notably underrepresented. This is particularly evident for topics like 'Ethical Framework' and 'Socioeconomic Risks',

[5] The key questions considered for the construction of the relevant metrics are the following: (a) Has the country established a process for meaningful public participation in the development of a national AI Policy?, (b) are materials about the country's AI policies and practices readily available to the public?, (c) Do the following goals appear in the national AI policy: "Fairness," "Accountability," "Transparency," "Rule of Law," "Fundamental Rights"?, (d) Has the country by law established a right to Algorithmic Transparency?

[6] For more information refer to: https://oecd.ai/en/incidents?search_terms=%5B%5D&and_condition=false&from_date=2014-01-01&to_date=2024-05-12&properties_config=%7B%22principles%22:%5B%5D,%22industries%22:%5B%5D,%22harm_types%22:%5B%5D,%22harm_levels%22:%5B%5D,%22harmed_entities%22:%5B%5D%7D&only_threats=false&order_by=date&num_results=20, and here: https://oecd.ai/en/catalogue/metrics

[7] For further details see: https://incidentdatabase.ai/

which are either assigned disproportionately low weights in indices, such as the Global AI Index (GAI), completely omitted from metrics like the Government AI Readiness Index (GAIRI) and the Global Cities AI Readiness Index (GCAIRI), or inadequately addressed due to deficient methodological frameworks in indices like the Artificial Intelligence and Democratic Values Index (AIDVI). This indicates a need for more robust and comprehensive approaches in the development and application of AI metrics that adequately reflect these critical dimensions.

Other topics related to 'Automation', 'Collaboration', 'National Challenges', and 'National Security', are absent from global AI metrics. While AI-related automation has made significant strides, the full extent of its development, particularly in complex and unpredictable environments, has yet to be realized. As regards 'Collaboration', such indices often prioritize quantifiable data that can be uniformly measured and compared across countries, which makes the qualitative aspects of international collaborations challenging to include in a standardized global indicator/ranking format. The omission of 'National Challenges' perceived as AI-enabled solutions for societal issues (i.e., healthcare, agriculture, education) in global AI metrics is indeed a significant lack. Given the direct relevance of these challenges to the Sustainable Development Goals (SDGs), there is a pressing need to develop indices that can specifically measure AI's impact in these areas. Such metrics would not only demonstrate how AI contributes to solving societal issues but also align AI developments with broader global objectives that genuinely benefit humanity. 'National Security' is presumably excluded due to the sensitive nature of defense-related data, which countries might not wish to disclose publicly.

The analysis highlights a critical issue in the dynamics of policy development: the over-reliance on traditional metrics that are easy to measure rather than those that address deeper, more meaningful objectives. This tendency to prioritize measurable and immediate outcomes often skews policies toward visible economic and technological benchmarks, neglecting social outcomes. As a result, policies risk becoming disproportionately shaped by what existing metrics can capture, creating a reinforcing cycle that prioritizes areas already well-represented in traditional indicators while leaving critical dimensions underexplored.

Traditional metrics in science and technology further exemplify this limitation, often falling short in capturing the complexities and transformative potential of AI. For instance, conventional indicators such as the number of AI-related scientific publications or patents may no longer adequately reflect a country's performance and innovative capacity. This limitation becomes even more pronounced with the rise of generative AI, which has unprecedented potential in knowledge creation—spanning from authoring scientific papers to generating entirely new forms of knowledge (Feuerriegel et al. 2024; Goto and Katanoda 2023). These capabilities challenge foundational concepts, such as transparency, originality, authenticity, and accountability (Hamed and Wu 2024; Picano 2024), raising profound questions about the evolving dynamics between human and artificial roles. These challenges become even more significant when extrapolated in critical domains like education, healthcare, and defense, wherein the interplay between human autonomy, control, and oversight with AI dynamics, carries far-reaching ethical, practical and vital implications (Hadlington et al. 2024; Farina et al. 2024). Therefore, addressing AI's proper integration into society necessitates a paradigm shift, one that embraces these complexities and acknowledges the distinctive capabilities and transformative impact of AI.

A promising approach to addressing the complexities of identifying, measuring, and monitoring AI's societal impact is to actively engage with society to gain a deeper understanding of its implications. This can be achieved by creating dedicated platforms that systematically capture, analyze, and monitor AI's influence on critical socio-economic sectors. In view of this, initiatives like the "AI Incidents Monitor" (developed by the OECD, accessible at oecd.ai/incidents), the "AI Incident Database" (developed by Partnership on AI, accessible at incidentdatabase.ai), the "AI, Algorithmic, and Automation Incident and Controversy Repository" (developed by the University of California, Berkeley, and accessible at aiaaic.org), and the "MITRE AI Incidents Repository" (developed by MITRE Corporation, mitre.org) exemplify efforts already underway. These repositories offer valuable insights into real-world AI incidents and hazards, providing a strong foundation for developing metrics that assess AI's societal and sector-specific impacts across domains like healthcare, education, agriculture, defense, etc. while also illuminating national/localization priorities and unique integration challenges faced by individual countries.

Considering the existential risks posed by AI, such approaches are essential for evaluating its impact on advancing or potentially hindering progress toward the Sustainable Development Goals (SDGs). They provide an evidence base for scrutinizing key issues including transparency (e.g., ensuring AI algorithms in healthcare diagnostics are interpretable and auditable), ethical challenges (e.g., addressing biases in AI models used in education that could reinforce existing inequalities), and risks (e.g., the environmental impact of training large-scale AI models). In this context, the role of civil society and public in general is pivotal. Public engagement enriches the construction of socially conscious metrics and informs policy development, ensuring inclusivity and responsiveness to societal needs. The growing recognition of society's role in AI governance has naturally expanded to influence the development of regulatory frameworks. Perspectives like experimental and anticipatory regulation reflect this shift, embedding public participation and engagement as fundamental components of the regulatory process (Nesta 2024; CUP 2024).

This approach would disrupt the existing cycle by ensuring a balanced representation of diverse topics within global AI metrics while fostering a more holistic, inclusive, and ethically grounded integration of AI, effectively translating value-driven social and ethical considerations into actionable policies (Schiff 2023; Foffano et al. 2023).

In conclusion, there is a significant space for international bodies and national policymakers to refine AI evaluation frameworks to better reflect a comprehensive view of AI's impacts. This would entail a strategic recalibration of AI metrics to encompass a broader spectrum of considerations, ensuring that AI development is aligned with both global standards and local societal needs. This recalibration is crucial for promoting an AI future that is inclusive, ethical, and fully integrated into the fabric of global society.

4 Contribution

This study contributes to the understanding of the interplay between AI metrics and policymaking by addressing the conceptual alignment between how AI progress is measured and how it is framed in national policies. To our knowledge, this is the first research effort to explore the intersection of global AI metrics and National AI Strategies (NAIS), providing both empirical and theoretical insights. The analysis reveals a critical misalignment between the global measurement of AI progress and the policy objectives articulated in national strategies, highlighting the need for more nuanced and comprehensive evaluation frameworks.

Theoretically, this study advances the understanding of AI governance by offering insights into bridging the gap between policy intentions and measurable outcomes. Through critical analysis, it contextualizes the insights of this research within the broader premises of studies focusing on NAIS priorities (Papyshev and Yarime 2023; Saheb and Saheb 2023) and those examining global AI metrics (Erkkilä 2023). In doing so, it introduces a framework for understanding how national AI priorities can inform the creation of balanced, context-sensitive metrics. This contribution enriches ongoing debates about the transparency and accountability in data-driven AI governance, emphasizing the necessity of responsible metrics that align with societal needs and ethical imperatives (Papyshev and Yarime 2024; Schiff 2023; Charles et al. 2022).

On an empirical level, this research identifies and examines 15 topics across 43 NAIS documents, uncovering national variations in strategic priorities. This study provides a comprehensive dataset that highlights both the gaps and overlaps between national strategies and global AI metrics. While not the primary focus of this research, the dataset offers opportunities for future studies to delve deeper into individual countries' specific priorities. Such analyses could reveal patterns in how certain strategic topics correlate with specific metric outcomes, providing valuable insights for refining national AI evaluation frameworks.

Third, this research provides actionable insights for policymakers and metric developers by advocating for the integration of societal dimensions and public engagement into AI evaluation frameworks (Foffano et al. 2023; Erkkilä 2023; Wilson and Van Der Velden 2022). It underscores that AI metrics should not only focus on technical and economic outputs but also reflect the broader societal priorities outlined in NAIS. To this end, the development of specialized platforms for systematically capturing, analyzing, and assessing AI's impact on critical socio-economic sectors would offer substantial value. Moreover, it lays the theoretical foundation for developing country-specific metrics that align more closely with national priorities, enabling governments to monitor the effectiveness of AI policies and measure their impact on societal challenges, such as healthcare, education, and sustainability.

5 Appendix

See Table 3 and Fig. 3.

Table 3 Mapping of NAIS topics to aligned AI Indices and data sources

Topic	Aligned Indices	Reflection in Indices	Data Sources
Human Capital	GAI, GCAIRI, GAIRI	Strong focus across all indices, emphasizing workforce education, talent attraction, and STEM training. However, diversity and inclusivity metrics are underweighted with respect to other metrics	LinkedIn, UNESCO, Kaggle, World Bank, Mercer, OECD, UN data
Infrastructure	GAI, GCAIRI, GAIRI	Well-covered, focusing on technological readiness like high-performance computing and data centers. However, rural infrastructure and equitable distribution of AI benefits receive little attention	ISP Review, Top500, Speedtest, OECD reports, World Economic Forum, UN e-Government Survey
Research and Development (R&D)	GAI, GCAIRI, GAIRI	Strongly emphasized, focusing on academia-industry collaboration and innovation outputs. However, the societal relevance and ethical dimensions of R&D activities are rarely incorporated in these indices	SCOPUS, World Bank, UNESCO, Times Higher Education
Innovation Strategy	GAI, GCAIRI, GAIRI	Well covered with emphasis on pilot projects, innovation hubs, and government support. However, the indices overlook sustainability and long-term societal impact of innovation beyond economic benefits	National AI strategies, World Bank, GovTech Maturity Index, Times Higher Education, SCOPUS
Private Sector	GAI, GCAIRI, GAIRI	Evaluation of corporate readiness, innovation, and investments. Ethical considerations and labor market implications of private sector AI activities are minimally covered	Tortoise Media, Oxford Insights, Oliver Wyman Forum
Socioeconomic Risks	AIDVI, GAI	Weakly addressed overall, with sporadic focus on inequality, public trust, and job displacement. Metrics prioritize economic growth, leaving gaps in evaluating societal risks posed by AI adoption	Ipsos MORI, Mercer, UN data, Kaggle, OECD
Regulation	GAI, GAIRI	Regulation is primarily measured in terms of AI governance and legal frameworks. Ethical governance metrics are underemphasized or lack actionable insights	UNCTAD, Global Data Barometer, ISO standardization efforts, Worldwide Governance Indicators
Public Sector	GCAIRI, GAIRI	Moderately reflected, focusing on AI adoption in government services. However, metrics lack depth in assessing the quality and inclusiveness of AI-driven public services or public awareness initiatives	World Bank, UN e-Government Survey, OECD AI Policy Observatory
Ethical Framework	AIDVI, GAI	Weakly addressed in AIDVI through broad questions about AI principles and human rights. GAI focuses narrowly on regulatory frameworks rather than practical societal impacts of ethical AI	OECD/G20 AI Principles, UNESCO Ethics Recommendation, Universal Declaration of Human Rights
Data Governance	GAI, GAIRI	Focuses on data availability and privacy laws but lacks nuanced coverage of data quality, bias mitigation, and equitable data access for marginalized communities	OECD Open Data Charter, ISP Review, National data privacy legislation sources
Alignment with Transnational Organizations	AIDVI	AIDVI highlights alignment with global frameworks (OECD, UNESCO), but this is limited to binary metrics of endorsement. Implementation and actual adherence to these principles are inadequately assessed	OECD/G20 AI Principles, UNESCO Ethics Recommendation, UN SDG reports
Collaboration	-	-	-
National Challenges	-	-	-
Automation	-	-	-
National Security	-	-	-

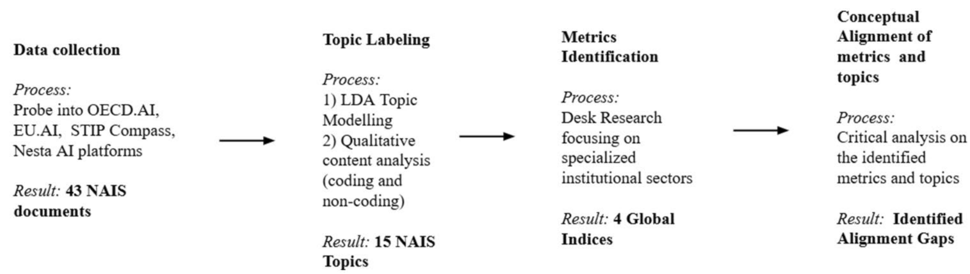

Fig. 3 Methodology flowchart showing the sequential steps from data collection to critically analyzing metrics and aligning topics

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Data availability The data and code used for this study are available at: https://github.com/baudsiou/AI-Metrics-and-Policymaking

Declarations

Conflict of Interest The authors have no competing interests.

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